Genetics Notes

Course Notebook

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1 Chapter 0 introduction

1.0.1 A brief history of modern genetics

- Early history:
  - Inheritance of acquired characteristics(Lamarck)
  - Blending inheritance
  - Natural Selection
    Darwin failed to answer 2 questions
  - Mendel
  - Morgan
  - McClintock

1.0.2 What is Genetics

heredity, inheritance variation

relationship between heredity and variation:

1.0.3 Why do we study Genetics

preductive and preventative medicine gene therapy: butterfly child–insertion OTC deficiency–adeno-associated virus(without insert into genome) ARCUS 《the code breaker》

1.0.4 the themes of modern genetics

regulatory networks modular - DNA is the fundamental information molecule of life

human genome has about 3*10^3 base pairs

why DNA is the heredity molecule: have the ability to store, replicate, mutate, express information RNA is less stabla than DNA(脱氧和 U?)

2 Chapter 1 Mendal's Law of Inheritance

2.1 Mendel's breakthrough

> Key words - law of segregation（分离定律）- law of independent assortment（独立分配定律）- reciprocal cross（反交）- backcross（回交）- testcross（测交）- inbreeding, incross（内交，近交）- outbreeding, outcross（外交，远交）- P: parental generation（亲本，亲代）- F1: first filial generation（子一代）- F2: second filial generation（子二代）- selfing line（自交系）- inbred line, inbred strain（近交系）- product rule（乘法法则，

> 相乘定律）- sum rule（加法法则，相加定律）- hybrid（杂种）- monohybrid（单杂种，单因子杂种）- dihybrid （双因子杂种，双因子杂合子）- multihybrid（多基因杂种）- Punnett square（Punnett 棋盘式方格）- branched-line diagram（分支图）- pedigree（系谱，家谱）- consanguineous mating（近亲交配）

Garden pea(*Pisum sativum*): - Stigma & Anthers -> **self-fertilization** - **Pure breeding**: produce offspring carring specidic parental traits that remain constant from generation and generation

TOPIC: What is an ideal experimental organism > Aging and longetivity research: Shortest-lived vertebrate species bred in the lab: *Turquoise killifish* > (due to the short rain season in Africa) > - Cancer and aging research: Naked mole rat > (Rarely get cancer; resistant to some types of pain; survive up to 18 minutes without oxygenl; Do not age; Fertility does not drop as they age)

Keys to Success: 1. ideal experimental organism - Vigorously(茁壮的) grow - Self-fertilize - Easy to cross-fertilize - Produce large number of offspring each generation 2. pure-breeding lines (2-year trial experiment) 3. analyzed traits with descrete alternative form(Black and white experiment) 4. numerical analysis

2.1.0.1 Law of Segregation

Monohybrid crosses reveal units of inheritance and **Law of Segregation** (**Reciprocal cross** 互交)

Two alleles for each trait separate (**segregate**) during gamete formation, and then unite at random, one from each parent, at fertilization.

> Using The Punnett Square to present the law of segregation

对于实验的若干解释: **Dominant** & **recessive** - Disappearance of traits in F1 generation and reappearanc in the F2 generation **disproves the hypothesis that traits blend** - 实验要求: Trait must have two forms that can each breed true - One form must be hidden when plants with each trait are interbred -

Testcross reveals unkown genotype: to cross with a homozygous recessive

2.1.0.2 Law of Independent Assortment

Dihybrid crosses reveal the **law of independent assortment**

对于实验的若干解释：- A dihybrid is an individual that is heterozygous at two genes - F2 generation contained both parental types and recombinant types - Alleles of genes assort independently, and can thus appear in any combination in the offspring

During gamete formation different pairs of alleles segregate independently of each other

2.1.0.3 Summary

- Inheritance is **particulate**, not blending

- There are two copies of each trait in a germ cell
- Gametes contain one copy of the trait
- Alleles (different forms of the trait) segregate randomly
- Alleles are dominant or recessive - thus the difference between genotype and phenotype
- Different traits assort independently

2.1.0.4 Mendelian inheritance in HUMAN Most traits in humans are due to the interaction of multiple genes and do not show a simple Mendelian pattern of inheritance

Pedigrees help us infer if a trait is from a single gene and if the trait is dominant or recessive

Pedigree analysis

- A vertical pattern of inheritance indicates a rare dominant trait > Huntington's disease: A late-onset neurodegenerative genetic disorder that causes uncontrolled movements, emotional problems, and loss of cognition. > **Poly Q** - a gain of toxic function of mHtt

- A horizontal pattern of inheritance indicates a rare recessive trait > Cystic fibrosis: A fatal and incurable disorder in which the lungs, pancreases, intestines and other organs become clogged with a viscous mucus that interfere with breathing and digestion

- Puzzling pedigree > Progeria: a *de novo* dominant trait, an extremely rare genetic disorder with premature aging symptoms there is no known cure, few people with progeria exceed 13 years of age. > > > > Mutation occurs during cell division in a newly conceived zygote or in the gametes of one of the parents > > Treacher Collins syndrome(TCS): a *de novo* dominant trait, a disorder of craniofacial development.

2.2 Extension to Mendal's Law

key word - Incomplete dominance - Codominance - Multiple alleles - Pleiotropy - lethality - complementary gene action - complementation test - heterogeneity - redundancy, redundant gene - epistasis, epistatic effect, epistatic gene - hypostatic gene - recessive epistasis - dominant epistasis - penetrance（外显率）__incomplete penetrance - expressivity（表现度）__variable expressivity - discrete trait（不连续性状）- quantitative trait（数量性状）- continuous trait（连续性状）- phenocopy（拟表型，表型模拟）

2.2.0.1 Single-gene inheritance Incomplete dominance:

F1 hybrids that differ from both parents express an intermediate phenotype. Neither allele is dominant or recessive to the other. Phenotypic ratios are same as genotypic ratios.

> Snapdragons *Antirrhinum majus*
>
> horse

Codominant:

both traits appear equally, Phenotypic ratios are same as genotypic ratios.

> lentil coat patterns
>
> blood type

Multiple alleles:

1. Genes may have multiple alleles that segregate in populations.
2. Although there may be many alleles in a population, each individual carries only 2 of the alternatives.
3. Dominance or recessiveness is always relative to a second allele.

> ABO blood type: Enzymes add sugar A or sugar B to a base consisting of a sugar polymer known as substance H, resulting in antigen A or antigen B respectively
>
> The eye color of Drosophila

- Establishing dominance relations between multiple alleles: Perform **reciprocal crosses** between pure breeding lines of all phenotypes.

TOPIC:Mutations are the source of new alleles Allele frequency is the percentage of the total number of gene copies represented by one allele. Wild-type allele ——the most common allele in a population Mutant allele ——a rare allele in the same population Monomorphic ——gene with only one common, wild-type allele Polymorphic ——gene with more than one common allele

Pleiotropy:

a single gene determines more than one distinct and seemingly unrelate characteristic, Each protein can have a cascade of effects on an organism, Some alleles may cause lethality

> Respiratory problems & Sterile

> A. Yellow must be AYA, and AY is dominant to A B. AY is recessive lethal! AYAY die in utero and do not show up as progeny.

> Sickle-Cell Anemia: both Mutliple alleles & Pleiotropy Pleiotropy part: Sickling & Resistance to malaria & Recessive lethality - clog the small blood vessels −Shortness of breath, muscle cramps, fatigue - Very fragile −Anemia
>
> Malaria parasites cause sickle-shaped blood cells to break down before the malaria parasites can reproduce.

Some alleles may cause **lethality**

2.2.0.2 Multifactorial inheritance

The action of two or more genes

Penetrance and expressivity:

Each genotypic class (defined in terms of the presence or absence of the dominant allele of two genes) determines a particular phenotype: - both present - one present - the other present - neither present

> The shape of cockscombs

Complementary gene action:

> color in sweet peas: A_B_(purplr) & others(white) = 9:7

> A possible explaination

> in human deafness Complementation: if offspring receiving two mutations one from each parent - express the wild-type phenotype, complementation has occurred. Complementation test can determine if mutations arise from the same or different genes.

9:7 ratio is a phenotypic signature of complementary gene interaction where dominant alleles of two genes act together (A_B_) to produce a trait while other three genotypic classes do not.

Epistasis:

a gene interaction in which the effects of an allele at one gene is masked, inhibited, or modified by one or more other genes, known as modifier genes

> coat color in Labrador retriever dog
>
> **recessive epistasis**

9:3:4 is a telltale ratio of recessive epistasis at F2

> Bombay blood type the woman lack of substance H(hh),so the recessive allele(h) masks the effects of any ABO alleles
>
> Bombay blood type can only accept blood from another Bombay blood type indivitual

> summer squash
>
> Green(Chlorophylls)->Yellow(Carotenoid)
>
> **Dominant epistasis**

> chicken feather: A profuces color only in absence of B

12:3:1 and 13:3 are telltale ratios for dominant epistasis.

Duplicate effect:

provide alternative genetic determination of a specific phenotype.

> Duplicate effect in shepherd's purse
>
> **Redundancy**: more than one gene acting in a totally same pathway At this time we can do gain-function experiment(overexpress), and do the lost-function experiment to know the genotype however sometimes is not redundancy but complementary⋯.? *test*

Testing two gene and one gene hypothesis

- 9:4:3(recessive epistasis) & 1:2:1(incomplete dominance)-> backcross
- pedigree analysis can be used to tes trait inheritance hypothesis in humans: > OCA(Oculocutaneous albinism)

Interactions between genes and the environment

- Penetrance: percentage of a population with a particular genotype that show the expected phenotype > Penetrance can be complete (100%) or incomplete (e.g.,retinoblastoma penetrance is 75%). **Gene interactions**: two genes can interact to determine one trait

- Expressivity: degree or intensity with which a particular genotype is expressed in a phenotype > Expressivity can be variable or unvarying.

Factors that alter the phenotypic expression of genotype include: - Modifier genes - Modifier genes have a more subtle, secondary effect, while major genes have a large influence. - The environment > Hydrangea （绣球花）: > The more alkaline the soil, the pinker the flowers > Siamese cat: > Melanin is produced only in the cooler extremities

> Phocomelia: chemical exposure **Phenocopy**: a change in phenotype arising from environmental agents that mimics the effect of a mutation at a gene - Chance

3 Chapter 2 The Chromosome Theory of Inheritance and Linkage Analysis

3.1 The chromosome theory of inheritance

key words: - somatic cell（体细胞）- germ cell（种质细胞）- homologous chromosome（同源染色体）- autosome（常染色体）- crossing-over, crossover（交换）- chiasma (chiasmata)（交叉）- recombination nodule（重组结）- karyotype（核型分析）- hemizygous（半合子）- sex-limited trait（限性性状）- sex-influenced trait（从性性状）- criss-cross inheritance（交叉遗传）

3.1.0.1 Chromosomes: the carriers of genes

Genes reside in the nucleus

> microscopists studying fertilization in frogs and sea urchins

> Flemming investigated the process of cell division He identified **chromatin** was correlated to threadlike structures in the cell nucleus –the chromosomes

> Cajal

> Walter Sutton Studied **Great lubber grasshopper**–Smotic cells contained 22 chromosomes +XY; Gametes contained 11 chromosomes and X/Y easy to distinguish different pairs of chromosome

The boveri-sutton chromosome theory: - chromosomes as the carriers of genetic material. - linear structure–loci - paired factors - carriers of genetic material

Haploid gametes produce diploid zygotes at fertilization > **Haploid**: a single chromosome set > **Diploid**: two matching chroomosome sets > Anatomy of a chromosome

Homologous chromosomes match in size, shape, and banding patterns

- centromere
- Metacentric Chromosomes
- Acrocentric Chromosomes
- Metaphase chromosomes are identified by the position of chromosomes

Karyotype(核型分析): shows the stained chromosomes of a single organism in homologous pairs of **decreasing size**(大小递减的排列)

TOPIC: WHY DOES THE NUMBER OF CHROMOSOMES MATTER - Only one: Restrain the diversity - One hundred: mistakes in cell division

Nonhomologous chromosomes are qualitatively different

> Jimsonweed （蔓陀罗）: trisomy （三体）in any of these 12 chromosomes produced a unique phenotype of seed pod shape, deviating from a wild-type spherical shape.
>
> Trisomy 21 why only trisomy 21 patients: for other chromosomes maybe lethal, they are bigger and have more genes.

3.1.0.2 Sex chromosome and sex determination Sex chromosome: - Provide basis for sex determination - One sex has matching pair（同配性别）XX - Other sex has one of each type of chromosome（异配性别）XY

SRY(sex determining region of Y) is the primary determinant of maleness > in sex-reversed XX male, one of the X chromosomes carries SRY > sex-reversed XY females lacks functional SRY on Y

- a little piece of DNA that makes girls boys: Sox9(downstream of SRY, in chronosomes 11) determines maleness SRY->Sox9->Maleness

- Dosage compensation(how to proof?) > mammals: inactivation of one X > flies: 2X transcription rate of a single X > worms: repression of both X to half

- Sex determination in fruit flies and humans
  - human: the presence or absence of Y, and human can tolar the XXX
  - Drosophila: the number of X chromosomes

> Natural sex changes in clownfish When the dominant female dies, the dominant male changes sex to take her place, while the largest juvenile male becomes the new dominant male, triggered by social cues and hormonal shifts 推测有可能和感受群体中的信息素有关

3.1.0.3 Mitosis, meiosis and gametogenesis

Somatic cells vs germ cells

Mitosis: Genetically conservative - Prophase-Prometaphase-Metaphase-Anaphase-Telophase-Cytokinesis - Interphase: the period between cell divisions, consisting of the G1, S, and G2 phases - S phase: chromosomes replicate to form sister chromatids during synthesis;

Meiosis:One source of the genetic variation that fuels evolution - crossing-over: An exchange of parts between nonsister chromatids during pachytene. As a result, chromatids may no longer be of purely maternal or parternal origin. - contributes to genetic diversity: - Independent assortment of nonhomologous chromosomes creates different combinations of alleles among chromosomes - crossing-over between homologous chromosomes creates different combinations of maternal and paternal information. - Random union of genetically distinct sperm and egg.

- **Oogenesis**
  - Diploid germ cells called oogonia(卵原细胞) multiply by mitosis to produce primary oocytes
  - Oogenesis begin in the fetus(婴儿期)

  > Maternal age correlates with meiotic segregational errors

  > Prenatal genetic diagnosis:
  > - amniocentesis
  > - non-invasive sequencing of cell-free genome NIPT: obtain the fetal DNA in a sample of mother's blood

- **Spermatogenesis**
  - Symmetrical meiotic divisions produce four functional sperm
  - Mitosis produces diploid primary spermatocytes
  - Spematids mature into functional sperm

3.1.0.4 Validation of the chromosome theory of inheritance

Morgen's theory:

TOPIC: Nondisjunction of chromosomes during meiosis Calvin Bridges's Primary exceptional progeny

Validation:

3.1.0.5 Sex-linked traits in human

> X-linked recessive trait in human–hemophilia difficulty in which bloof can't clot normally
> treatment: injection of cloting factot

X-linked recessive trsits exhibit 5 characteristics: - Trait appears in more males than females - Mutation and trait never pass from father to son - Affected male pass X-linked mutation to all daughters, who are heterozygous carriers（交叉遗传）- The trait often skips a generation（隔代遗传）- Trait only appears in successive generations if sister of an **affected male** is a carrier. If so, one half of her sons will show the trait

Sex-limited traits Sex-influenced traits: traits can show up in both sexes, but are expressed differently in each due to hormonal differences.

> Premature pattern baldness（早老性斑秃）: dominant in men; recessive in women

3.2 Linkage and recombination

key words: - linkage（连锁）- recombination（重组）- parental type（亲组型）- intrachromosomal recombination（染色体内重组）- interchromosomal recombination（染色体间重组）- genetic marker（遗传标记）- linkage group（连锁群）- goodness of fit（适合度）- chi square test（卡方检验）- null hypothesis（零假说，无效假说）- p value（p 值）- recombination frequency（RF，重组率）- centimorgan（cM，厘摩）; map unit（m.u.，图距单位）- Locus（基因座）

3.2.0.1 Gene linkage and recombination

Autosomal genes can also exhibit linkage

> Colorblindness and hemophilia are X-linked traits

Detect linkage by generating a double heterozygote and crossing to homozygous recessive (testcross)

Recombination frequency, RF:

RF = Recombinant number / (Parental number + Recombinant number) > A recombination frequency of less than 50% is a diagnostic for linkage.

3.2.0.2 The Chi-square test and linkage analysis

Chi-square test measures "goodness of fit" between observed and expected (predicted) results

Chi-square test can reject the null hypothesis, but it cannot prove a hypothesis - Null hypothesis: observed values are not different from the expected values (Expect a 1:1:1:1 ratio of gametes) - Alternative hypothesis: observed values are different from expected values (Expect significant deviation from 1:1:1:1 ratio)

Data needed: - Total number of progeny - How many classes of progeny - Number of offspring observed in each class

Degrees of freedom (df)–df = N −1 (where N is the number of classes)

Determine a p value using chi-square value and df

Use p value of 0.05 as cutoff Chi-square values that lie in the yellow region of this table allow rejection of the null hypothesis with >95% confidence

3.2.0.3 Recombinstion: A result of crossing-over

Chiasmata mark the sites of recombination

Morgen's theory: Different gene pairs exhibit different linkage rates because genes are arranged in a line along a chromosome. The closer together the two genes are on the chromosome, the less their chance of being separated by an event that cuts and recombines the line of genes

3.2.0.4 Summary

- Genes close together on the same chromosome are linked and do not segregate independently
- Linked genes lead to a larger number of parental class than expected in double heterozygotes cross
- Mechanism of recombination is crossing-over
- Chiasmata are the visible signs of crossing over
- Recombination frequencies reflect physical distance between genes
- Recombination frequencies between two genes vary from **0% to 50%**

3.3 Mapping: locating genes along a chronosome

key word: - two-point cross（两点杂交）- two-point testcross（两点测交）- three-point cross（三点杂交）- three-point testcross（三点测交）- coefficient of coincidence（C，并发系数）- interference（I，干涉）- chromosomal interference（染色体干涉）- I = 1 - C - gene mapping（基因定位，基因作图）- tetrad analysis（四分子分析）- octad（八分子）- parental ditype（PD，亲二型）- nonparental ditype（NPD，非亲二型）- tetratype（T，四型）- twin spot（孪生斑）- genetic mosaic（遗传镶嵌）

3.3.0.1 Two point crossess

Recombination frequencies reflect the distances between two genes

1 RF = 1 m.u. (map unit) = 1 cM (centiMorgan)

limitation: - Difficult to determine gene order if two genes are close together - Actual distances between genes do not always add up - Pairwise crosses are time and labor consuming

3.3.0.2 Three point crosses

Correction for Double Crossovers (DCO)

The RF of linked genes cannot exceed 50%: - Meioses without crossovers produce only parental chromosomes. - Single and double crossovers produce a 1:1 parental to recombinant chromosome ratio on average.

Double recombinants indicate order of three genes: double cross is in the middle

Chromosomal interference: Occurrence of one crossover reduces likelihood that another crossover will occur in adjacent parts of the chromosome

Measuring interference: - coefficient of coincidence =

$$\frac{\text{Observed Double Crossovers (DCO)}}{\text{Expected Double Crossovers (DCO)}}$$

- interference = 1 - coefficient of coincidence

Summary: - Cross true breeding mutant with wild-type - Analyze F2 individuals (**males** if sex linked) - Determine order of genes based on parentals and recombinants - Determine genetic distance between each pair of recombinants - Calculate coefficient of coincidence and interference

> Cystic fibrosis (CF): A recessive condition 1. Pedigrees -> CF is encoded by a single gene; 2. Gene PON encoding Serum enzyme is linked to CF; 3. Linkage analyses with hundreds of DNA markers -> CF is on the long arm of chromosome 7; 4. Fine mapping with several other markers; 5. In 1992, CF gene was cloned.

DNA marker: a piece of DNA of known size, representing a specific locus, that comes in identifiable variations. These allelic variations segregate according to Mendel's laws.

> a typical marker: RFLP–Restriction fragment length polymorphism southern blotting

> Nomenclature for Drosophila genes - Wild-type allele - denoted with a "+"- Mutant allele - denoted with no extra symbol - Recessive mutation - gene symbol is in lower case - Dominant mutation - gene symbol is in upper case

linkage provides the potential for transmitting favorable combinations genes Recombination produces great flexibility in generating new combination of alleles.

DNA polymorphisms: - Most polymorphisms do not influence phenotype - Four categories of genetic variation: - SNP - DIP/InDel: Insertion and Deletion - SSR: Simple Sequence Repeat - CNV: Copy Number Variation

Copy number variant: 顾名思义

Single Nucleotide Polymorphism

Genotyping can identify carriers and homozygous individuals. many SNPs only in introns, so it has small influence in phenotype > sickle cell anemia is caused by a SNP in the Hb gene

> Neurofibromatosis: - autosomal, dominantly inherited - positional cloning example determines whether or nota SNP is linked to the neurofibromatosis gene，但有的时候会出现重组，利用重组率计算基因距离

Determing genotype by sequencing PCR products.

Simple Sequence Repeat are highly polymorphic Genotype is dertermined by PCR at many SSR loci diferent from person to person

Multiplex PCR is used for DNA fingerprinting: > match DNA from crime scene to a person

> Multiolex PCR for DNA fingerprinting

3.3.0.3 Tetred analysis in fungi ideal experimental organism−Tetrad - all four haploid products of meiosis are contained in ascus - Ascospores（子囊孢子）within ascus germinate into haploid individuals - Phenotype of haploid fungi is direct representation of their genotype

- Meiosis can generate three kinds of tetrads: PD & NPD (parental ditype & nonparental ditype) 这是自由组合的事情
- Meiosis can generate three kinds of tetrads: T(tetratype)

Linkage is demonstrated by PDs outnumbering NPDs: PD/NPD > 1, linked; PD/NPD = 1, unlinked

A disproven model: Recombination before chromosome duplication, which predicts most tetrads containing recombinant spores would be NPDs instead of Ts.

3.4 Mitotic recombination and genetic mosaics

3.4.0.1 Mitotic recombination produces genetic mosaics twin spots, single spot, singed spot - yellow (y) mutant −yellow body - Wild type (y+) −brown body - singed (sn) mutant −short and curled bristles - Wild type (sn+) −long and straight bristles

3.4.0.2 Application of mitotic recombination Application: > Mosaic clonal analysis using the FLP-FRT system 相当于强制进行有丝分裂重组 > >

Origin of cancer > retinoblastoma > 可能有很多不同的生病原因 > > > > heterozygous form homozygous cells

> **The two-hit model**: It takes two mutational events to cause the cancer. The first mutational event can be inherited, whereas the second "hit"results in the loss of the remaining normal allele (gene).

mitotic recombination: start from heterozygous so the alleles must be dominant

3.5 Mechanism of homologous recombination and gene conversion

key words: - abnormal segregation（异常分离）- gene conversion（基因转变，基因转换）- chromatid conversion（染色单体转变）- half-chromatid conversion（半染色单体转变）- post-meiotic segregation（减数后分离）- homologous recombination（同源重组，HR）- general recombination（一般重组）- hybrid DNA model（杂合 DNA 模型）- double-strand break model（双链断裂模型）- breakage and rejoining（断裂与重接）- heteroduplex DNA（异源双链 DNA）- hybrid DNA（杂合 DNA，杂种 DNA）- branch migration（分支迁移）- Holliday intermediate（Holliday 中间体）- displacement loop（D-loop）

3.5.0.1 Homologous recombination Types of DNA recombination: - Homologous recombination - Site-specific recombination - Transposition 转座

> Meselson and Weigle: Experimental evidence

- homologous recombination: 同源重组
  - The double stranded break model:
  - Step 1: Double-strand break formation
  - Step 2: Resection
  - Step 3: Strand invasion
  - Step 4: Formation of double Holliday junction
  - Step 5: Branch migration
  - 会形成一个 Heteroduplex region 区域

  > Holliday model

- site-specific recombination
  - The FLP-FRT system
  - Cre-Lox recombination
  - Brainbow strategies

3.5.0.2 Gene conversion one ellele is converted to the allele on the homologous chromosome, resulting in nonreciprocal recombination

Gene conversion vs. mutation: - It occurs at a higher frequency (0.1 to 1%) than the corresponding mutation events; - It always produces the allele present on the homologous chromosome, not a new allele; - It correlated about 50% of the time with reciprocal recombination of flanking markers.

> 侧翼标记 想要观察中间的 G 基因是否发生了突变或重组，通过 A（上游）和 B（下游）这两个特征来“圈定”这段区域。

4 Chapter 3 The Concepts of Gene and Mutation

4.1 The Conception of Gene

4.1.0.1 DNA as the genetic material

> Griffith's demonstration of bacterial transformation

> Avery's experiment

> Alfred Hershey and Martha Chase

4.1.0.2 A gene is a discrete linear set of nucleotide pairs Complementation test can determine if mutations arise from the same or different genes.

How recombination within a gene could generate a wild-type allele?

> Benzer's experiment: T4 phage - Can examine a large number of progeny to detect rare mutation events (100-1000 progeny/hour) - Phase chromatin is "naked", allowing much higher recombination frequency - Could allow only recombinant phage to proliferate while parental phages died
>
> Step 1: complementation test (trans) -> find out all mutations in rll-A
>
> Complementation test is *trans* combination, testing whether this 2 mutations are in same gene or not Control group is *cis* combination, testing whether the mutation is dominant or recessive
>
> Step 2: recombination test -> between rll-A mutations
>
> Recombination test: testing whether recombination appears in the same gene Control group: exlude the possible of revertants

Gene structure: - A gene consists of different parts that can each mutate; - Recombination between different mutable sites in the same gene can generate a normal, wild-type allele; - A gene performs its normal function only if all of its components are wild type.

- The minimal recombination frequency: 0.02 cM;
- T4 Chromosome: 1.8×105 bp, 1500 cM (120 bp/cM);
- 0.02 cM (0.02/1500) 1.8×10^5 = 2.4 bp
- Gene is not an indivisible entity
- Different nucleotide pairs within a gene are independently mutable, and recombination can occur between nucleotide pairs within a gene as well as between genes

4.1.0.3 The central dogma

> The human dystrophin gene: RNA splicing

RNA splicing: - there's specific splice site - Alternative RNA splicing - dendritic self-avoidance

Translation 部分省略了

4.1.0.4 Transposable elements –the jumping genes

Activator transposase catalyzes excision and integration

Transposition: The movement of small segments of DNA – entities known as transposable elements (TEs) – from one position in the genome to another - Retroposons: transpose via reverse transcription of an RNA intermediate 先变成 RNA 再逆转录为 DNA 然后才插入 - LINEs - SINEs - The Alu element and human diseases - Alu insertions are sometimes disruptive and can result in inherited disorders. - HERVs

- LTR: long terminal repeats
- Transposons move their DNA directly

Transposon structure: Transposase enzyme recognizes inverted repeats and excises the transpon 1. P element in original genomic position and Transposase 2. P element excised 3. Excision of P element leaves a gap at its original location 4. Repair of gap using a sister chromatid or homologous chromosome containing a P element Transposon remains in original position 5. Repair of gap using a homologous chromosome lacking a P element Transposon no longer at original position 6. Repair by Nonhomologous end joining(NHEJ)

An inserted element is flanked by a short repeat: transposition footprint

> hybrid dysgenesis of *drosophila*

4.2 Functional dissection of a gene through mutation

4.2.0.1 Mutation

Mutation: Changes in base sequences that modify the information content of DNA.

Classification of Mutations:

By effect on structure - Small-scale mutations - Point mutation - Insertion - Deletion - Large-scale mutations - Amplification - Deletion - Translocation - Inversion

By effect on function - Loss of function - Gain of function - Dominant negative - Reverse mutation

By inheritance - Inheritable germline mutation - Non inheritable somatic mutation

Different genes, different mutation rates - Average mutation rate in gamete-producing eukaryotes is higher than that of prokaryotes - Diploid organisms can tolerate more mutations than haploid organisms - Human sperms have higher mutation rate than human eggs - average:

Mutation rate = 1 X 10-8 - Each child contains about 60 mutations (base pairs different from either parent), do not effect phenotype - Mutation rate in sperm = 2-4 X 10-8 - Revertants are more rare than forward mutations

The molecular basis of spontaneous mutations

> fluctuation test Hypothesis 1：Resistance is a physiological response to a bactericide Hypothesis 2：Resistance arises from random mutation
>
> Each colony represents the descendants of a single resistant bacterium

> Replica plating verifies that bacterial resistance is the result of preexisting mutations

- Bacterial resistance arises from mutations that occurred before exposure to bactericide
- Mutations occur as the result of random processes

point mutation: ：A type of mutation that causes the replacement of a single base nucleotide with another nucleotide of the genetic material, DNA or RNA. - Transition: purine replaced by another purine, or pyrimidine replaced by another pyrimidine - Transversion: purine replaced by a pyrimidine, or pyrimidine replaced by a purine

insertion and deletion: - block of 1 or more bp lost from DNA - block of 1 or more bp added to DNA

Natural processes: x-ray; UV light; Oxidative damage - Depurination: DNA 中的 嘌呤（adenine A 或 guanine G）碱基 从糖-磷酸骨架上脱落，形成 apurinic site (AP site)(1000/hr in every cell) - Deamination of C : C changed to U; Normal C-G -> A-T after replication

Errors in replication: tautomerization

unequal crossing-over: > Unequal crossing-over between red and green photoreceptor genes can change gene number and create hybrid photoreceptor proteins > > >

Transposable elements:

Unstable trinucleotide repeats - CGG repeat: 会导致 non-coding region 的甲基化水平上升，影响转录 - Anticipation: become apparent at an earlier age as it is passed on to the next generation. In most cases, an increase of severity of symptoms is also noted

- slipped mispairing

- 有可能的治疗措施: dCas-Tet1(demethylation), 但问题是 target to other CGG repeats > Fragile X syndrome: The trinucleotide repeat is present within the non-coding region, the repeat expansion could affect the expression of the gene. > Huntington's disease: The trinucleotide repeat is present within the protein-coding region, the repeat expansion leads to production of a mutant toxic protein with gain of function.

The molecular basis of induced mutations

base analogs: chemical structure almost identical to normal base (5BU)

hydroxylating agents: Hydroxylating agents add an −OH group alkylating agents: Alkylating agents add ethyl or methyl groups > Ethyl Methanesulfonate > dosage 很重要，matching & lethal

deaminating agents: Deaminating agents remove amine (-NH2) groups

intercalating agents: > Profalvin > 平面芳香族化合物 > Proflavin 插入两条 DNA 链之间, 使 DNA 复制时发生 frameshift mutation（移码突变）

> Ames test to screen for chemicals that cause mutations and, therefore, might cause cancer.
>
> Rice liver enzyme: 模拟哺乳动物体内的代谢环境, 将化学物质转化为活性代谢物

The phenotypic consequences of mutations

- Wild type
- Base substitutions
- Frameshift mutation
- Point mutation in non-coding sequence

Silent mutations do not alter the amino acid sequence

Missense mutations replace one amino acid with another - Conservative −*chemical properties* of mutant amino acid are similar to the original amino acid - Nonconservative −*chemical properties* of mutant amino acid are different from original amino acid

Nonsense mutations change codon that encodes an amino acid to a stop codon (UGA, UAG, or UAA)

Frameshift mutations result from insertion or deletion of nucleotides with the coding region that change the reading frame for the reminder of the translation process. No frameshift if multiples of three are inserted or deleted

Mutations outside the coding sequence can disrupt gene expression: - Mutations in the sequence of a promoter -> diminish or prevent transcription - Changes in a splice-acceptor or donor site -> no mature mRNA or splicing errors - Mutations affecting ribosome binding site -> diminish the efficiency of translation - Changes in sites that regulate stability or localization of the mRNA -> instability or mislocalization of mRNA and protein

Null (amorphic) mutations: completely block function of a gene product (e.g. deletion of an entire gene)

Hypomorphic mutations −gene product has weak, but detectable, activity

Some loss-of-function alleles can show incomplete dominance

Rare dominant loss-of-function allele > 通常 loss-of-function 突变是隐性的，因为一个正常拷贝可以弥补功能（haplosufficient）> 显性 LoF 突变发生在某些情况下：> - Haploinsufficiency（半量不足）：phenotype is sensitive to gene dosage (i.e. 50% of gene product) > 见下

Hypermophic mutation: generate more gene product or the same amount of a more efficient gene product > FGFR3 gene > Achondroplasia (a form of short-limbed dwarfism; 先天性软

骨发育不全) is caused by a dominant hypermorphic allele of FGFR3. The altered protein is always active.

Neomorphic mutations –generate gene product with new function or that is expressed at inappropriate time or place

Antimorphic(dominant negative): Mutant subunits block the activity of normal subunits kinase 的问题：就是 block ATP 的结合位点; 或者 block 专门的激酶活性位点

Biological repair system

Proofreading decreases mistakes made during replication

DNA repair systems:

accurate repair system: Particularly important for removing uracil - Base excision repair removes damaged bases(BER) - DNA glycosylases remove altered nitrogenous base. - Nearby nucleotides removed - New DNA systhesized to fill gap

- Nucleotide excision repair corrects damaged nucleotides(NER)
  - UvrA-UvrB complex scans for distortions to double helix (e.g. thymine dimers)
  - UvrB-UvrC complex nicks the damaged DNA
  - DNA polymerase fills in the gap
- homologous and nonhomologous repair(HR)
- Nonhomologous end-joining(NHEJ)

Error-prone repair systems: - SOS system: - Adds random nucleotides opposite damaged bases - Microhomology-mediated end-joining (MMEJ) - Similar to NHEJ, except nucleotides are removed at doublestranded breaks leading to small deletions

Defects in DNA repair lead to human diseases:

> Xeroderma pigmentosum (XP) An autosomal recessive genetic disorder of DNA repair in which the ability to repair damage caused by ultraviolet (UV) light is deficient

> BRCA1 and breast cancer BRCA1 (breast cancer 1, early onset) is human tumor suppressor gene. It is involved in double-strand break repair by homologous recombination and contribute to susceptibility to breast cancer.

4.2.0.2 Organelle genetics Mitochondrial DNA is circular in shape while nuclear DNA is linear in shape.

Nuclear and organellar genomes cooperate with one another, semiautonomous

- Heteroplasmic cells contain a mixture of organelle genomes
- Homoplasmic cells contain only one type of organelle genome

Maternal inheritance of Xenopus mtDNA > 用 mtDNA 做亲子鉴定的原因 : short lots of copies > The mutation rate of animal mtDNA is higher than that of nuclear DNA > Usually no change in mtDNA from parent to offspring

4.2.0.3 mtDNA mutations and human health

- Tissues with higher energy requirements are less tolerant of mutant mitochondria
- Tissues with low energy requirements are affected only when the proportion of wild-type mitochondria is greatly reduced

Three-person in vitro fertilization, two ways: